

HDOC – Day 24

C Programming Assignment

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Batch 15

Date - 17/09/2026

Problem Requirements

Display a menu of five increasing-pattern options. Use a switch statement to select the user's choice. Each pattern must be implemented in a separate parameterless function with no return value (void). For each activity, the algorithm and test cases are provided below, followed by one complete C program.

1. Increasing right-angled triangle

Algorithm

1. Start.
2. Input number of rows n.
3. For i = 1 to n, print i asterisks using an inner loop.
4. Move to the next line after each row.
5. Stop.

Test Input	Expected Output	Result
n=4	* * * * * * * * * *	Pass
n=1	*	Pass

2. Increasing row-number triangle

Algorithm

1. Start.
2. Input number of rows n.
3. For each row i from 1 to n, print the value i exactly i times.

4. Move to the next line after each row.
5. Stop.

Test Input	Expected Output	Result
n=4	1 2 2 3 3 3 4 4 4 4	Pass
n=2	1 2 2	Pass

3. Increasing natural-number pattern

Algorithm

1. Start.
2. Input number of rows n.
3. For each row i from 1 to n, print numbers j = 1 through i.
4. Move to the next line after each row.
5. Stop.

Test Input	Expected Output	Result
n=4	1 1 2 1 2 3 1 2 3 4	Pass
n=3	1 1 2 1 2 3	Pass

4. Consecutive-number triangle

Algorithm

1. Start.
2. Input number of rows n and set num = 1.
3. For each row i from 1 to n, print i values of num; increment num after every print.
4. Move to the next line after each row.
5. Stop.

Test Input	Expected Output	Result
n=4	1 2 3 4 5 6 7 8 9 10	Pass
n=3	1 2 3 4 5 6	Pass

5. Increasing alphabet pattern

Algorithm

1. Start.
2. Input number of rows n.
3. If n is outside 1 to 26, display an error and return.
4. For each row i from 1 to n, print characters from A through A+i-1.
5. Move to the next line after each row.
6. Stop.

Test Input	Expected Output	Result
n=4	A A B A B C A B C D	Pass
n=26	Rows increase from A up to A ... Z	Pass
n=27	Invalid input. Enter rows from 1 to 26.	Pass

Complete C Program

```
#include <stdio.h>

void starTriangle(void);
void rowNumberTriangle(void);
void naturalNumberPattern(void);
void consecutiveNumberTriangle(void);
void alphabetPattern(void);
int main(void)
{
    int choice;
    printf("INCREASING PATTERNS MENU\n");
```

```

printf("1. Increasing right-angled triangle\n");
printf("2. Increasing row-number triangle\n");
printf("3. Increasing natural-number pattern\n");
printf("4. Consecutive-number triangle\n");
printf("5. Increasing alphabet pattern\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice) {
case 1: starTriangle(); break;
case 2: rowNumberTriangle(); break;
case 3: naturalNumberPattern(); break;
case 4: consecutiveNumberTriangle(); break;
case 5: alphabetPattern(); break;
default: printf("Invalid choice.\n");
}
return 0;
}

void starTriangle(void)
{
int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0) { printf("Invalid input.\n"); return; }
for (i = 1; i <= n; i++) {
for (j = 1; j <= i; j++) printf("* ");
printf("\n");
}
}

void rowNumberTriangle(void)
{
int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0) { printf("Invalid input.\n"); return; }
for (i = 1; i <= n; i++) {

```

```

for (j = 1; j <= i; j++) printf("%d ", i);
printf("\n");
}
}

void naturalNumberPattern(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0) { printf("Invalid input.\n"); return; }
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}

void consecutiveNumberTriangle(void)
{
    int n, i, j, num = 1;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0) { printf("Invalid input.\n"); return; }
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("%d ", num++);
        printf("\n");
    }
}

void alphabetPattern(void)
{
    int n, i, j;
    printf("Enter number of rows (1-26): ");
    scanf("%d", &n);
    if (n < 1 || n > 26) {
        printf("Invalid input. Enter rows from 1 to 26.\n");
        return;
    }
}

```

```

for (i = 1; i <= n; i++) {
for (j = 0; j < i; j++) printf("%c ", 'A' + j);
printf("\n");
}
}

```

Compilation ,Execution & Testing

```

Session Actions Edit View Help
(kali㉿kali)-[~]
$ vi hdoc24.c

(kali㉿kali)-[~]
$ cc hdoc24.c
hdoc24.c:1:1: warning: function 'consecutiveNumberTriangle' declared but not defined [-Wmissing-declarations]
1
^
(kali㉿kali)-[~]
$ ./a.out
1: Page 4
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 1
Enter number of rows: 4
*
* * hdoc24.c
* * *
* * * *
* * * * hdoc24.c

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 1
Enter number of rows: 1
1
*

```

```

(kali㉿kali)-[~]
$ ./a.out
1: Page 4
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 2
Enter number of rows: 2
1
2 2

```

```

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 4
1
1 2 3 4
1 2 3
1 2 3 4

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 3
Enter number of rows: 3
1
1 2 3
1 2 3
1 2 3

```

```

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 4
1
2 3
4 5 6
7 8 9 10

(kali㉿kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 4
Enter number of rows: 3
1
2 3
4 5 6

```

```

(kali@kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter number of rows (1-26): 4
A
A B
A B C
A B C D

```

```

Session Actions Edit View Help
Enter number of rows (1-26): 26
A
A B
A B C
A B C D
A B C D E
A B C D E F
A B C D E F G
A B C D E F G H
A B C D E F G H I
A B C D E F G H I J
A B C D E F G H I J K
A B C D E F G H I J K L
A B C D E F G H I J K L M
A B C D E F G H I J K L M N
A B C D E F G H I J K L M N O
A B C D E F G H I J K L M N O P
A B C D E F G H I J K L M N O P Q
A B C D E F G H I J K L M N O P Q R
A B C D E F G H I J K L M N O P Q R S
A B C D E F G H I J K L M N O P Q R S T
A B C D E F G H I J K L M N O P Q R S T U
A B C D E F G H I J K L M N O P Q R S T U V
A B C D E F G H I J K L M N O P Q R S T U V W
A B C D E F G H I J K L M N O P Q R S T U V W X
A B C D E F G H I J K L M N O P Q R S T U V W X Y
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Increasing natural-number pattern
(kali@kali)-[~]
$ ./a.out
INCREASING PATTERNS MENU
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice: 5
Enter number of rows (1-26): 27
Invalid input. Enter rows from 1 to 26.

```